

A sharp spectral-gap refinement for the switch chain

Tom Ayre
ayre.tom at gmail.com

October 7, 2026

Status. Conditional research note prepared with assistance from Codex. This note retains the triple-resampling inequality and slice equality-projection inequality of OpenAI's *Polynomial Mixing of the Switch Chain for Every Graphical Degree Sequence* (September 25, 2026). The new step strengthens its nonnegative disjoint-pair estimate to $D \succeq H$. We also use the exact Johnson-slice gap. A targeted literature search did not locate this statement; publication priority is not established. The source was read at the archived source directory at revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Separate agent reviews of the source's finite row, triple, slice and disjoint-cell arguments found no gap. These are local AI-assisted checks, not formal verification or independent external human peer review.

Let Ω_d be the simple labeled graphs with fixed graphical degree sequence d on $n \geq 4$ vertices. For a vertex pair a , let E_a be conditional uniform resampling of its singleton neighbors, $h_a = I - E_a$, and $H = \sum_a h_a$. Let L be the unnormalized Laplacian whose states are Ω_d and whose edges are single valid switches, each neighbor counted once. Set $D = \sum_{a \cap b = \emptyset} h_a h_b$, summing ordered pairs. All inner products use the uniform probability measure.

Theorem 1 (Conditional candidate). *Assume the source's triple inequality $H_T^2 \succeq H_T$ and slice inequality $S \preceq G$ described below. Then*

$$D \succeq H, \quad H^2 \succeq 2H, \quad H \preceq L.$$

If $|\Omega_d| > 1$, then $\text{gap}(L) \geq 2$, with sharp universal constant. For the source's lazy four-vertex-proposal chain,

$$\text{gap}(P_d) \geq \frac{1}{6 \binom{n}{4}}, \quad t_{\text{mix}}(\varepsilon) \leq \left\lceil 6 \binom{n}{4} \left(\frac{1}{2} \log |\Omega_d| + \log \frac{1}{2\varepsilon} \right) \right\rceil \quad (0 < \varepsilon < 1/2).$$

Thus $t_{\text{mix}}(1/4) = O(n^6)$ rather than the displayed $O(n^8)$ bound. The spectral-gap bound is attained for $d = (2, 2, 1, 1, 0, \dots, 0)$. The nonlazy uniform-pair heat-bath chain $K = \binom{n}{2}^{-1} \sum_a E_a$ has sharp universal gap $\text{gap}(K) \geq 2/\binom{n}{2}$.

Proposal conventions. The chain P_d holds with probability $1/2$. Otherwise it chooses a four-vertex set uniformly, chooses uniformly one of the six ordered pairs of distinct perfect matchings on that set, and replaces the first matching by the second if all removed edges are present and all added edges are absent; otherwise it holds. Thus every distinct switch neighbor has transition probability $1/[12 \binom{n}{4}]$. The chain K chooses a pair uniformly and samples uniformly from its entire pair fiber, including its current state. This chain is a convex combination of orthogonal projections, so its eigenvalues are nonnegative.

Borrowed inputs. For disjoint pairs a, b , let $g_{a,b} = (I - T_{a,b})/2$, where $T_{a,b}$ switches the four cross edges when possible. On an a -fiber, $R_{a,b}$ projects onto the centered indicator that the two singleton neighbors in b receive the same endpoint. It is zero if either is not a singleton, or if the indicator is constant. On a uniform q -subset slice of N singleton labels put

$$G = \frac{1}{2} \sum_{k < l} (I - \tau_{kl}), \quad S = \sum_{k < l} Q_{kl},$$

where Q_{kl} is the equality-indicator projection. Source Lemma `sl:comparison` proves $S \preceq G$. The elementary Johnson calculation in the appendix gives $G \succeq (N/2)(I - E)$ on nonconstant slices. Source Proposition `tri:gap` proves $H_T^2 \succeq H_T$. The disjoint-cell decomposition used next is spelled out in its `sections/disjoint.tex`.

A sharper disjoint-cell estimate. Fix disjoint pairs a, b . In a source disjoint cell, outside singleton sets have sizes A, B ; the 2×2 cross matrix has total s , first row sum r and first column sum c . The counts of compatible outside subsets are $a_r = \binom{A}{k+1-r}$ and $b_c = \binom{B}{l+1-c}$. In the dependent case $s = 2$, the matrix-multiplicity table has center two, four adjacent entries one, and zero corners. If both off-center events have positive mass, $a_1 b_1 > 0$. Set

$$\alpha = \frac{a_0 + a_2}{a_1}, \quad \beta = \frac{b_0 + b_2}{b_1}, \quad \rho^2 = \frac{\alpha\beta}{(2+\alpha)(2+\beta)}.$$

Here ρ is the magnitude of the rank-one correlation in the source. For $0 \leq k \leq A$,

$$\alpha = \frac{A-k}{k+1} + \frac{k}{A-k+1} \leq (A-k) + k = A,$$

and similarly $\beta \leq B$. Thus, for $\kappa_A = A/(A+2)$ and $\kappa_B = B/(B+2)$, we have $\rho^2 \leq \kappa_A \kappa_B$.

The source's correlation-plane component $f = xu + yv$ satisfies $h_a f = y(v + \rho u)$, $h_b f = x(u + \rho v)$; each parenthesized vector has squared norm $1 - \rho^2$ and their mutual product is $\rho(1 - \rho^2)$. Weighted Young gives

$$2xy\rho(1 - \rho^2) \geq -\kappa_A y^2(1 - \rho^2) - \kappa_B x^2(1 - \rho^2).$$

The first error line lies in $\text{ran } R_{a,b}$ and the second in $\text{ran } R_{b,a}$, as in the source. The common orthogonal component contributes at least $2\|g_{a,b}f\|^2$. Therefore

$$2\langle h_a f, h_b f \rangle \geq 2\|g_{a,b}f\|^2 - \kappa_A \|R_{a,b}f\|^2 - \kappa_B \|R_{b,a}f\|^2. \quad (*)$$

Independent cells satisfy this without the error terms. The coefficients have a fiberwise interpretation: $R_{a,b} = 0$ unless $c = 1$, and on such fibers the singleton count is $N = A + 2$. Its coefficient is therefore $1 - 2/N$; symmetrically for $R_{b,a}$. Constant slices have $h_a = 0$.

Summing the positive remainder. Sum $(*)$ over unordered disjoint pairs. For fixed a , the resulting operator on a nonconstant fiber is

$$G - (1 - 2/N)S \succeq (2/N)G \succeq I - E.$$

The first step uses $S \preceq G$ and $N \geq 2$, the second the Johnson gap. Averaging and summing over a gives $D \succeq H$.

The source’s exact identity is

$$H^2 = \sum_{T \in \binom{[n]}{3}} H_T^2 - (n-3)H + D.$$

The triple inequality now gives $H^2 \succeq 2H$. Connectivity by switches identifies its kernel with constants, hence $2\text{Var}(f) \leq \langle f, Hf \rangle$. Let $J_a = \sum_{k < l} (I - \tau_{kl})$ on each fiber. Its nonzero gap is N , so $h_a \preceq J_a/N \preceq J_a/2$. Each switch has exactly two pair representations; thus $\sum_a J_a = 2L$, and $H \preceq L$. This proves $\text{gap}(L) \geq 2$. The source normalization is $I - P_d = L/[12\binom{n}{4}]$; the usual L^2 -to-TV estimate gives the stated mixing bound.

Sharpness. For $d = (2, 2, 1, 1)$ the only realizations have edge sets $\{12, 13, 24\}$ and $\{12, 14, 23\}$. They are joined by one switch; the Laplacian has spectrum $\{0, 2\}$. Adding isolated vertices preserves this state space. The lazy proposal kernel has off-diagonal entry $1/[12\binom{n}{4}]$, so its gap is exactly $1/[6\binom{n}{4}]$. Exactly two vertex pairs have nonconstant fibers in this example. Each such resampling is averaging on the two states, so H has spectrum $\{0, 2\}$ and $\text{gap}(K) = 2/\binom{n}{2}$ as well.

Checks and limitations. The operator diagnostic checks $D \succeq H$, $H^2 \succeq 2H$, $H \preceq L$ and the nontrivial spectral gaps for all 144 degree-sequence classes on $n = 4, 5, 6$, scanning 33,856 labeled graphs. A separate implementation constructs L directly from switches, without using pair fibers or the operators H, D . It checks all 486 degree-sequence classes on $n = 4, 5, 6, 7$, including 366 nontrivial classes. The minimum observed gap is 2, within tolerance 10^{-7} . These are finite floating-point diagnostics, not proofs. The source’s triple and equality-projection inequalities were reviewed locally in separate Codex passes without locating a gap. External peer review and formal verification remain outstanding; no fresh Lean kernel verification was performed. No definitive novelty claim is made. Only the spectral-gap constant is asserted sharp, not the $O(n^6)$ mixing exponent.

Primary literature checked. The closest located sharp analogue is Fu, Qin and Wang, *Spectral Gap for the Binary Fixed-Margin Swap Chain*, <https://arxiv.org/abs/2606.22636>. It treats binary fixed-margin matrices, equivalently bipartite graphs, rather than the general undirected state space considered here. It also supplies the three-row inequality used in the source framework. Johnson-spectrum switch/Curveball comparison is already standard: <https://arxiv.org/abs/1709.07290> and <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.APPROX-RANDOM.2018.36>. For undirected Curveball and the two pair representations per switch, see <https://pmc.ncbi.nlm.nih.gov/articles/PMC6072652/>. For the unified switch-chain analysis see <https://arxiv.org/abs/1903.06600>. The potentially new part is the sharper disjoint-interaction estimate and sharp universal spectral-gap consequence within the source framework. A targeted search found no matching general undirected statement; it does not rule out a known implicit consequence or establish publication priority.

A An elementary Johnson-slice gap proof

Let $\mathcal{X}_{N,q}$ be the uniform q -subsets of $[N]$, with $1 \leq q \leq N-1$, and let $J = \sum_{i < j} (I - \tau_{ij})$. For $A \subseteq [N]$ put $x_A = \prod_{i \in A} x_i$. Let V_k be the span of the monomials of degree at most k on the slice. For $|A| = k$, direct summation over transpositions with one endpoint in A gives

$$Jx_A = k(N - k + 1)x_A - (q - k + 1) \sum_{a \in A} x_{A \setminus \{a\}}. \quad (\text{J})$$

Indeed before using $\sum_i x_i = q$ the expression is $k(N - k)x_A - \sum_{a \in A} x_{A \setminus \{a\}} \sum_{b \notin A} x_b$, and each product in the second sum equals $(q - k + 1)x_{A \setminus \{a\}} - x_A$. Thus J preserves V_k , and modulo V_{k-1} acts as $\lambda_k = k(N - k + 1)$. Self-adjointness shows that on $V_k \cap V_{k-1}^\perp$ it acts by exactly this scalar: the difference $(J - \lambda_k I)f$ lies in both V_{k-1} and its orthogonal complement. Finally V_q is the full function space, because degree- q monomials are point indicators. On each nonconstant nonzero level,

$$\lambda_k - N = (k - 1)(N - k) \geq 0.$$

This proves $J \succeq N(I - E)$, or $G = J/2 \succeq (N/2)(I - E)$. The levels $q = 0, N$ are constant and need no gap estimate.